# Monitoring Deforestation and Land-Use Change in the Amazon Using Deep Learning on Satellite Imagery

Data Science II — Project Proposal | Rank: 1st Choice (Recommended)

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## Introduction

We propose to build a deep learning system that automatically tags satellite image tiles of the Amazon basin with the land-cover, land-use, and atmospheric conditions visible in them — including primary rainforest, selective logging, slash-and-burn agriculture, artisanal and conventional mining, roads, cultivation, and habitation. Because a single image tile can show several of these conditions at once (for example, "clear" weather over a region with both "primary" forest and "agriculture"), this is a **multi-label** image classification problem rather than a simple single-class task.

Our planned approach includes:

- **Data preprocessing:** Normalize the 4-band (Red, Green, Blue, Near-Infrared) GeoTIFF/JPEG image chips, and engineer an additional Normalized Difference Vegetation Index (NDVI) channel from the Red and NIR bands to make vegetation health and canopy loss more visible to the model. We will also examine the frequency of each of the 17 possible labels to identify severe class imbalance (e.g., "blow_down" and "artisanal_mine" appear in under 1% of tiles) and apply targeted oversampling and augmentation (rotations and flips, which are natural for top-down satellite imagery) for rare classes.
- **Model training:** Build a convolutional neural network for multi-label classification with a sigmoid output layer and binary cross-entropy loss per label. We will compare a CNN trained from scratch against transfer learning from an ImageNet-pretrained backbone (e.g., ResNet or EfficientNet) adapted to accept 4+ input channels, and will tune hyperparameters such as learning rate, thresholding per label, and class weighting.
- **Evaluation:** Measure performance with per-label and macro-averaged metrics, analyze which labels are hardest to detect (typically the rare, high-impact classes such as mining and logging), and generate confusion analysis and visualizations of model attention (Grad-CAM) over misclassified tiles.

**Related work:** This problem is based on the well-studied Kaggle "Planet: Understanding the Amazon from Space" competition, whose top solutions used ensembles of CNNs (e.g., ResNet/DenseNet variants) and achieved macro F2-scores above 0.93. More broadly, remote-sensing deep learning research (e.g., studies applying CNNs to Sentinel-2 imagery for land-cover classification) has shown that combining spectral indices like NDVI with CNN features improves detection of vegetation change compared to using raw RGB imagery alone. Our project will directly compare a raw-RGB CNN against an NDVI-augmented CNN to quantify this effect on our own imbalanced label set.

## Motivation

The Amazon rainforest loses tens of thousands of square kilometers of forest cover every year, and timely detection of illegal logging, mining, and land clearing is essential for environmental enforcement, carbon-accounting efforts, and climate policy. Manually reviewing satellite imagery at the scale required

to monitor an area the size of the Amazon basin is infeasible for human analysts; an automated multi-label tagging system could flag areas of concern far faster and more consistently, supporting conservation organizations and policymakers with actionable, near-real-time information.

From a learning perspective, this project pushes us beyond single-label image classification into multi-label prediction, multi-band (non-RGB) image processing, engineered domain-specific features (NDVI), and working with the kind of severe class imbalance that is common in real-world environmental and remote-sensing datasets. It is also a chance to connect a technical machine learning project to a concrete, high-stakes societal problem.

## Evaluation

**Success criteria:** We will consider the project successful if our best model achieves a macro-averaged F2-score of at least 0.85 across all 17 labels, and, more importantly, achieves a recall of at least 0.60 on the rare but high-impact classes (mining, selective logging, slash-and-burn agriculture) despite their scarcity in the training data. We also want to demonstrate a clear, measurable improvement from adding the engineered NDVI channel compared to an RGB-only baseline.

**Technical measurement:** We will use the F2-score (which weights recall more heavily than precision, matching the intuition that missing real deforestation is more costly than a false alarm) as our primary metric, both per-label and macro-averaged, alongside per-label precision/recall curves, Hamming loss, and confusion matrices. We will also perform qualitative error analysis using Grad-CAM heatmaps on misclassified tiles to understand whether errors stem from cloud cover, mixed land use within a single tile, or genuinely ambiguous visual patterns.

## Resources

**Dataset:** "Planet: Understanding the Amazon from Space" (Kaggle), which provides roughly 40,000 labeled 256×256 pixel satellite image chips (4-band, ~20 GB total) covering the Amazon basin, each tagged with one or more of 17 possible atmospheric and land-cover/land-use labels.

**Computational tools:** Google Colab (GPU runtime) for model training; Python with TensorFlow/Keras or PyTorch for building the CNN; rasterio and Pillow for reading multi-band imagery; scikit-learn for evaluation metrics; and matplotlib/seaborn for visualizing label distributions, training curves, and confusion matrices.

**Figure 1. Deforestation Detection Pipeline**




Figure 1. Planned end-to-end pipeline: multi-band satellite chips are preprocessed and augmented with an NDVI feature, passed through a CNN multi-label classifier, and evaluated with F2, precision, and recall.

## Contributions

**Team Members:** OluwaKoretimi Adebowale | Josselin Jimenez | Mehrab U Azan

- **OluwaKoretimi Adebowale — Data & Preprocessing:** Acquire and organize the Planet Amazon dataset, normalize the 4-band imagery, engineer the NDVI feature, analyze per-label frequency to quantify class imbalance, and implement the augmentation strategy (rotation/flip, oversampling of rare classes). *Writing:* draft the Introduction and Resources sections.
- **Josselin Jimenez — Modeling:** Build and train the CNN multi-label classifier (both the from-scratch and transfer-learning/pretrained-backbone versions), tune hyperparameters (learning rate, per-label decision thresholds, class weights), and manage experiment tracking across model variants. *Writing:* draft the Motivation section and the technical "planned approach" portion of the Introduction.
- **Mehrab U Azan — Evaluation & Interpretability:** Compute per-label and macro-averaged F2-score, precision/recall, and Hamming loss; generate confusion matrices; and run the Grad-CAM error analysis comparing the RGB-only baseline against the NDVI-augmented model. *Writing:* draft the Evaluation section.

All three of us will collaborate jointly on interpreting the final results, compiling the written report, and reviewing/editing every section (including this Contributions section and the Works Cited) so the workload stays balanced and everyone has visibility into the full project, not just their own piece.

## Works Cited

Kaggle. "Planet: Understanding the Amazon from Space." kaggle.com/competitions/planet-understanding-the-amazon-from-space. Accessed 2026.

Selvaraj, A., et al. "Deep Learning Approaches for Land Cover and Land Use Classification Using Remote Sensing Imagery." Remote Sensing Applications, various issues, 2019–2023.

# Automated Skin Lesion Classification for Early Melanoma Screening Using Deep Learning

Data Science II — Project Proposal | Rank: 2nd Choice

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## Introduction

We propose to build a convolutional neural network that classifies dermoscopic images of pigmented skin lesions into one of seven diagnostic categories — including melanoma, melanocytic nevus, basal cell carcinoma, actinic keratosis, benign keratosis-like lesions, dermatofibroma, and vascular lesions — with the goal of supporting early melanoma screening.

Our planned approach includes:

- **Data preprocessing:** Resize and normalize dermoscopic images, apply hair removal/inpainting and contrast enhancement (common artifacts in dermoscopy images), and use data augmentation (rotation, flipping, color jitter) to address the strong class imbalance in the dataset (benign nevi vastly outnumber melanoma and other rare categories). We also plan to fuse the available patient metadata (age, sex, lesion location) with image features in a multimodal model.
- **Model training:** Fine-tune an ImageNet-pretrained CNN backbone (e.g., EfficientNetB0 or ResNet50) on the seven-class classification task, using class-weighted loss to counter imbalance, and compare performance with and without the metadata-fusion branch.
- **Evaluation:** Measure per-class sensitivity/specificity (with particular attention to melanoma recall, since missed melanomas are the most clinically costly error), and apply Grad-CAM to visualize which regions of each lesion image drove the model's prediction, checking whether these align with clinically relevant lesion features (irregular borders, color variation).

**Related work:** This project builds on Esteva et al. (2017, *Nature*), which showed that a CNN trained on a large set of clinical images could match board-certified dermatologists at classifying skin cancer, establishing the feasibility of CNN-based skin lesion diagnosis. Subsequent ISIC (International Skin Imaging Collaboration) challenge submissions have further refined class-imbalance handling and multimodal (image + metadata) approaches, which we plan to draw on directly for our own imbalance-mitigation strategy. We will also review more recent work on attention-based CNN architectures for dermoscopy, which report that combining segmentation-derived lesion masks with classification features can further improve sensitivity on rare malignant classes, and we will consider this as a possible extension if time allows.

## Motivation

Melanoma survival rates are highly dependent on how early the cancer is detected, yet access to dermatologists is limited in many regions, and visual diagnosis of pigmented lesions is difficult even for trained clinicians. An accurate, accessible automated screening tool could serve as a low-cost pre-screening aid, helping flag high-risk lesions for follow-up and potentially reducing diagnostic delay, especially in underserved areas.

From a learning standpoint, this project exposes us to the specific challenges of medical imaging machine learning: severe class imbalance where the rare class is also the most clinically important, the

high cost of false negatives compared to false positives, combining image and tabular metadata in a single model, and using interpretability tools (Grad-CAM) that are essential for building trust in any healthcare-adjacent ML system. Working through these issues will also give us practical experience with the kind of responsible-AI considerations — such as communicating model uncertainty and avoiding overstated clinical claims — that are increasingly expected of anyone building ML systems that touch health data.

## Evaluation

**Success criteria:** We will consider the project successful if the final model achieves at least 80% recall (sensitivity) specifically on the melanoma class, while maintaining a balanced accuracy of at least 75–80% across all seven diagnostic categories. We will also consider it a success if the metadata-fusion model measurably outperforms the image-only baseline.

**Technical measurement:** Balanced accuracy (to account for class imbalance), per-class sensitivity and specificity, macro-averaged F1-score, one-vs-rest ROC-AUC per class, and confusion matrices. We will also qualitatively validate Grad-CAM heatmaps against the visual regions dermatologists typically use for diagnosis (lesion border and pigment-network irregularities) as a secondary interpretability check.

## Resources

**Dataset:** HAM10000 ("Human Against Machine with 10000 training images"), available via Kaggle/the ISIC Archive, containing 10,015 dermoscopic images across seven diagnostic categories, along with a metadata CSV (age, sex, lesion localization) for each image (~2.7 GB total).

**Computational tools:** Google Colab (GPU runtime); TensorFlow/Keras with a pretrained EfficientNetB0/ResNet50 backbone; OpenCV for image preprocessing (hair removal, contrast enhancement); scikit-learn for evaluation metrics; and a Grad-CAM implementation (e.g., tf-keras-vis) for interpretability.

**Figure 1. Skin Lesion Classification Pipeline**

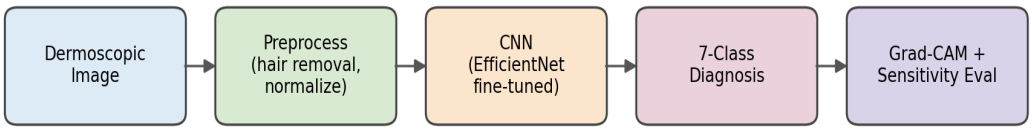



Figure 1. Planned pipeline: dermoscopic images are preprocessed, classified by a fine-tuned CNN into 7 diagnostic categories, and evaluated with sensitivity-focused metrics and Grad-CAM interpretability checks.

## Contributions

**Team Members:** OluwaKoretimi Adebowale | Josselin Jimenez | Mehrab U Azan

- **OluwaKoretimi Adebowale — Data & Preprocessing:** Acquire the HAM10000 images and metadata, perform hair removal/inpainting and contrast enhancement, and implement the class-weighted augmentation pipeline to address the imbalance between melanoma and the more common benign classes. *Writing:* draft the Introduction and Resources sections.

- **Josselin Jimenez — Modeling:** Fine-tune the pretrained CNN backbone (EfficientNetB0/ResNet50), build and integrate the metadata-fusion branch (age, sex, lesion location), and tune hyperparameters across the image-only and fusion model variants. *Writing:* draft the Motivation section and the technical "planned approach" portion of the Introduction.
- **Mehrab U Azan — Evaluation & Interpretability:** Compute per-class sensitivity/specificity, balanced accuracy, macro F1, and ROC-AUC; generate confusion matrices; and produce the Grad-CAM interpretability comparison against clinically relevant lesion regions. *Writing:* draft the Evaluation section.

All three of us will collaborate jointly on interpreting the final results, compiling the written report, and reviewing/editing every section (including this Contributions section and the Works Cited) so the workload stays balanced and everyone has visibility into the full project, not just their own piece.

## Works Cited

Esteva, Andre, et al. "Dermatologist-Level Classification of Skin Cancer with Deep Neural Networks." *Nature*, vol. 542, 2017, pp. 115–118.

Tschandl, Philipp, et al. "The HAM10000 Dataset, a Large Collection of Multi-Source Dermatoscopic Images of Common Pigmented Skin Lesions." *Scientific Data*, vol. 5, 2018.

# Detecting Signs of Depression in Social Media Posts Using NLP and Deep Learning

Data Science II — Project Proposal | Rank: 3rd Choice

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## Introduction

We propose to build a natural language processing system that classifies social media posts (e.g., Reddit posts) as showing signs of depression-related distress versus not, comparing a traditional machine learning baseline against a deep learning approach.

Our planned approach includes:

- **Data preprocessing:** Clean raw post text (removing URLs, usernames, markdown artifacts, and excessive punctuation), tokenize, and address class imbalance between depression-labeled and neutral/control posts through resampling and/or class-weighted loss.
- **Baseline model:** TF-IDF features combined with Logistic Regression and/or an SVM classifier, to establish a traditional ML baseline.
- **Deep learning model:** A BiLSTM over pretrained GloVe word embeddings, and a fine-tuned transformer (DistilBERT) to capture richer contextual language patterns, compared directly against the baseline.
- **Evaluation:** Compare all approaches on held-out test data, with particular attention to recall (missing an at-risk post is more costly than a false alarm), and perform an interpretability pass (top predictive n-grams for the baseline, attention weights for the transformer) to check whether the model is picking up on plausible linguistic markers of distress rather than spurious correlations.

**Related work:** This project is grounded in prior social-media mental health research, notably De Choudhury et al.'s studies on predicting depression risk from social media language patterns, as well as more recent transformer-based depression detection work on Reddit corpora, which report F1-scores in the 0.80–0.90 range using fine-tuned BERT-family models. We will use these results as a benchmark for our own model comparison, and will also review work on linguistic markers of depression (e.g., increased first-person singular pronoun use and absolutist language) to help us sanity-check that our models are learning meaningful signal rather than superficial dataset artifacts.

## Motivation

Depression is widely underdiagnosed, and many people express early signs of distress in their everyday online writing well before seeking (or being able to access) formal care. A system that can reliably flag concerning language patterns could support public-health researchers and platform moderation teams in directing at-risk users toward mental health resources, contributing to a broader societal goal of earlier, lower-barrier mental health support.

From a learning perspective, this project lets us build a complete NLP pipeline, compare classical ML against deep learning and transformer-based approaches on the same task, and grapple with the ethical considerations that come with any project involving sensitive, identity-adjacent text data — including using only publicly available, de-identified data intended for research use, and being careful about how

conclusions from the model are framed and communicated. This project also gives us a natural opportunity to practice model interpretability in an NLP setting, which is a different (and complementary) skill set from the image-focused interpretability techniques (e.g., Grad-CAM) commonly used in computer vision projects.

## Evaluation

**Success criteria:** We will consider the project successful if our best deep learning model achieves an F1-score of at least 0.80 on the held-out test set, and if it shows a clear, measurable improvement over the TF-IDF + Logistic Regression baseline — demonstrating that contextual language modeling adds real value over simple keyword-based approaches for this task.

**Technical measurement:** Accuracy, precision, recall, F1-score, and ROC-AUC for each model (baseline, BiLSTM, DistilBERT), confusion matrices to inspect the error trade-off between false positives and false negatives, and a qualitative interpretability comparison (top TF-IDF terms vs. transformer attention weights) to assess whether the higher-performing model is relying on clinically plausible language cues.

## Resources

**Dataset:** A Reddit-based depression-detection dataset available on Kaggle (e.g., a cleaned depression-vs-control subreddit post corpus, or the broader "Sentiment Analysis for Mental Health" dataset with tens of thousands of labeled statements across multiple mental health categories), used strictly as publicly available, de-identified data intended for research purposes.

**Computational tools:** Google Colab (GPU runtime) for transformer fine-tuning; Python with NLTK/spaCy for text preprocessing; scikit-learn for the TF-IDF baseline; PyTorch/TensorFlow and Hugging Face Transformers for the BiLSTM and DistilBERT models; and matplotlib/seaborn for visualizing metric comparisons across models.

**Figure 1. Mental Health Text Classification Pipeline**

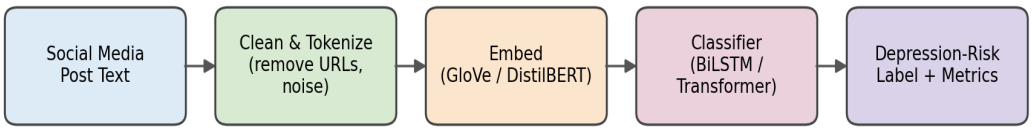



Figure 1. Planned pipeline: social media post text is cleaned and embedded, then classified by baseline and deep learning models, compared using recall-focused metrics.

## Contributions

**Team Members:** OluwaKoretimi Adebowale | Josselin Jimenez | Mehrab U Azan

- **OluwaKoretimi Adebowale — Data & Preprocessing:** Gather and clean the Reddit/text dataset (removing URLs, usernames, and noise), address class imbalance between depression-labeled and control posts, and build the TF-IDF feature representation used for the baseline model. *Writing:* draft the Introduction and Resources sections.

- **Josselin Jimenez — Modeling:** Build and train the BiLSTM-over-GloVe model and fine-tune the DistilBERT transformer, tune hyperparameters, and run the training experiments for both deep learning approaches. *Writing:* draft the Motivation section and the technical "planned approach" portion of the Introduction.
- **Mehrab U Azan — Evaluation & Interpretability:** Compute accuracy, precision, recall, F1, and ROC-AUC across the baseline, BiLSTM, and DistilBERT models; generate confusion matrices; and run the interpretability comparison (top TF-IDF terms vs. transformer attention weights). *Writing:* draft the Evaluation section.

All three of us will collaborate jointly on interpreting the final results, compiling the written report, and reviewing/editing every section (including this Contributions section and the Works Cited) so the workload stays balanced and everyone has visibility into the full project, not just their own piece.

## Works Cited


De Choudhury, Munmun, et al. "Predicting Depression via Social Media." *Proceedings of the International AAAI Conference on Web and Social Media*, vol. 7, no. 1, 2013.

Kaggle. "Sentiment Analysis for Mental Health." kaggle.com/datasets/suchintikasarkar/sentiment-analysis-for-mental-health. Accessed 2026.